

Information Extraction: NER

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Text Processing

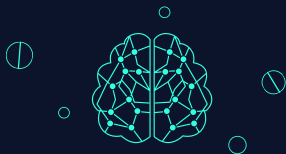
Department of Computer Science, University of Sheffield

Autumn Semester 2023

Diamond Lecture Theatre 5 (LT5)

6 PM

30 Nov.



RESPONSIBLE A.I.

THE ETHICAL & REGULATORY
CHALLENGES

An evening of talks & discussion on ethics and regulation in artificial intelligence research & development from experts in the field including:

- **Stephanie Hare:** author of *Technology is Not Neutral: A Short Guide to Technology Ethics* as well as broadcaster and researcher covering technology and politics
- **Dr. Jiahong Chen:** lecturer of law, researcher on the intersection between law and technology, and deputy director of EDI in the Law School at the University of Sheffield

Free for all students and
SLTSoc members



Tickets

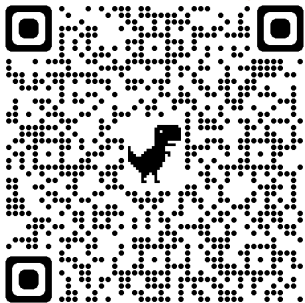


THE SLT SOCIETY



@thesltsociety

Deadline: 01/12 @ 5pm



<https://forms.gle/Ujm2kTjxLxQGc9f7>

CS Student Experience Video Competition

**Do you have something interesting to say about
your experience here in Sheffield?**

To celebrate the diversity in community, we particularly invite
students that are:

- Female / non-binary / LGBTQ+
- First of their family to attend University
- Part of the UoS Access+ programme

**Submit a short video for a chance to win
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Please use the QR code to complete your surveys.

(they take less than 10 minutes)



Learning objectives

- ▶ Explain NER
- ▶ Summarise knowledge engineering approaches for NER
- ▶ Summarise and apply supervised learning for NER
- ▶ Define the Entity Linking approach

Information Extraction: Overview

- ▶ **Introduction to Information Extraction**

- ▶ Definition + contrast with IR
- ▶ Example Applications
- ▶ Overview of Tasks
- ▶ Overview of Approaches
- ▶ **Evaluation + Shared Task Challenges**
- ▶ Briefest history of IE

- ▶ **Named Entity Recognition**

- ▶ Task
- ▶ Approaches: Rule-based, Supervised Learning
- ▶ Entity Linking

- ▶ **Relation Extraction**

- ▶ Task
- ▶ Approaches: Rule-based, Supervised Learning, Bootstrapping, Distant Supervision

Information Extraction: Evaluation

How to evaluate an IE system?

Needs a fully annotated corpus (**test**) → produced manually

- ▶ Correct answers, called **keys**, or **reference** or **ground truth**
- ▶ ideally: several annotations by several annotators → **interannotator agreement**

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Principal metrics are:

- ▶ **Precision** (how much of what a system returns is correct)
- ▶ **Recall** (how much of what is correct a system returns)
- ▶ **F-measure** (a weighted combination of precision and recall)

Information Extraction: Evaluation Metrics

Confusion matrix for a **binary classification** task →

Precision: how much of what system returns is correct

$$\text{precision} = P = \frac{TP}{TP + FP}$$

Recall: how much of what is correct system returns

$$\text{recall} = R = \frac{TP}{TP + FN}$$

F-measure: a weighted combination of precision and recall

$$F_{\beta} = \frac{(1 + \beta^2) * P * R}{\beta^2 * P + R} \Rightarrow F1 = \frac{2 * P * R}{P + R}$$

		REFERENCE		
		+	-	Σ
HYPOTHESIS	+	TP	FP	HP
	-	FN	TN	HN
Σ		RP	RN	N

Information Extraction: Evaluation Metrics

Confusion matrix for a **multiclass classification** task $\rightarrow$

Precision and **Recall** for class i :

$$precision_i = P_i = \frac{R_{i_H_i}}{H_i} \text{ and } recall_i = R_i = \frac{R_{i_H_i}}{R_i}$$

Micro and **macro** averages

$$\text{Micro-prec} = \frac{\sum_i R_{i_H_i}}{\sum_i H_i}$$

$$\text{Micro-recall} = \frac{\sum_i R_{i_H_i}}{\sum_i R_i}$$

$$\text{Macro-mes} = \frac{\sum_i mes_i}{\#classes} \text{ with } mes \in prec, recall$$

		REFERENCE			
		A	B	C	Σ
HYPOTHESIS	A	RA_HA	RB_HA	RC_HA	HA
	B	RA_HB	RB_HB	RC_HB	HB
	C	RA_HC	RB_HC	RC_HC	HC
Σ		RA	RB	RC	N

Shared Task challenges

Shared Task challenges or **evaluation campaigns**

competition among (international) research groups to build systems to address a specific task

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competition among (international) research groups to build systems to address a specific task

Conditions:

- ▶ clear **task definition**, annotated **text resources** for training, development and testing
- ▶ agreed **evaluation metrics** and **schedule**
- results are usually discussed during a conference

Shared task challenges in IE include: Message Understanding Conferences (**MUC**, 1985-1998), Automatic Content Extraction (**ACE**, 1999-2008), Text Analysis Conference (**TAC**, 2008-2018), **BioCreative** (molecular biology, 2004-2016)

⇒ **Define the core methodology of the field and have led to significant progress**

[Information Extraction]

Named Entity Recognition

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- ▶ **Named Entity Recognition**
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Information Extraction: NER - recap

Entity Extraction/Named Entity Recognition (NER)

Task: Identify the **extent** and the **type** of each textual mention of an entity

The set of types is determined in advance (e.g. organisation, person, date, etc...)

Cable and Wireless	today announced ...	Extent: 0-3 ; Type = ORG
IBM	and Microsoft	today announced ...
		Extent: 0-1 ; Type = ORG
		Extent: 2-3 ; Type = ORG
John Lewis	hired ...	Extent: 0-2 ; Type = ORG
Theresa May	hired.	Extent: 0-2 ; Type = PER

Information Extraction: NER - recap

Types of entities addressed by IE systems include:

► Named individuals

- Organisations (ORG), persons (PER), books, films, ships, restaurants ...
 - Cable and Wireless today announced ...; Extent: **0-3** ; Type = **ORG**
 - Barack Obama was the 44th president... ...; Extent: **0-2** ; Type = **PER**
- Geo-Political entities (GPE), locations (LOC)
 - The Mont Blanc intersects France, Italy and Switzerland. ; Extent: **1-3** ; Type = **LOC**
 - The Mont Blanc intersects France , Italy and Switzerland . ; Extent: **4-5** ; Type = **GPE**

► Times: temporal expressions dates, times of day

- Let's meet at 2pm next Friday ...; Extent: **3-4** ; Type = **TIME**
- Let's meet at 2pm next Friday ...; Extent: **5-6** ; Type = **DATE**

► Measures: monetary expressions, distances/sizes, weights ...

- This watch costs £35 ...; Extent: **3-4** ; Type = **MONEY**

Information Extraction: NER: coreference - recap

Coreference

Different textual expressions that refer to the same real world entity are said to **corefer**. **Coreference Task**: link together all textual references to the same **real world entity**,

Multiple references to the same entity in a text are rarely made using the same string:

- ▶ Pronouns: **Tony Blair** == **he**
- ▶ Names/definite descriptions: **Tony Blair** == **the Prime Minister**
- ▶ Abbreviated forms: **Theresa May** == **May**; **European Union** == **EU**
- ▶ Orthographic variants: **alpha helix** == **alpha-helix** == **α -helix** == **a-helix**

Can be seen as a separate task or as part of entity extraction task

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Information Extraction: Approaches to NER

Knowledge-engineering

- ▶ leverage linguistic resources created by experts

Supervised learning

- ▶ use of machine learning techniques

Bootstrapping

- ▶ Use of **seed patterns** to identify named entities
- ▶ Use known named entities to generate new patterns
- ▶ Rinse, repeat

Distant supervision / lightly supervised methods

- ▶ ~ bootstrapping a machine learning system

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NER: Knowledge Engineering Approaches

Dominant approach in the 1990s and still in use in many IE systems today.

Such systems typically use

- ▶ named entity **lexicons** and
- ▶ **manually authored** pattern/action rules or regular expression/FST recognisers

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Example: NER system, developed for participation in MUC-6

- ▶ described in Wakao et al. (1996) [Wakao et al., 1996]
- recognizes **organisation**, **person**, **location** and **time** expressions in **newswire texts**

Another system: **GATE Annie** [Cunningham et al., 2014]

→ Sheffield technology!

<https://cloud.gate.ac.uk/shopfront/displayItem/annie-named-entity-recognizer>

NER: Knowledge Engineering Approaches

System has four main stages:

1. Lexical processing
2. NE parsing
3. Discourse interpretation - Coreference Resolution
4. Discourse interpretation - Semantic Type Resolution

NER: Knowledge Engineering Approaches

1. Lexical processing

Rule-based NER systems use **specialized lexicons** → **gazetteers**
(= geographical directory)

The Wakao et al. system has specialised lexicons for:

- ▶ **Organisations** –
2600 entries
- ▶ **Locations** – **2200**
entries
- ▶ **Person names** –
500 entries
- ▶ **Company designators**: e.g. Corp,
Ltd – **94** entries
- ▶ **Person titles**: e.g. Mr, Dr, Reverend
– **160** entries

NER: Knowledge Engineering Approaches

1. Lexical processing

Rule-based NER systems use **specialized lexicons** → **gazetteers**
(= geographical directory)

Why not use even larger gazetteers?

- ▶ Gazetteer of British Place Names containing over 50,000 entries
- Many NEs occur in multiple categories
- **The larger the lexicons the greater the ambiguity**
- ▶ Ex.: Ford ⇒ **company** or **person** or **place**
- ▶ listing of names is never complete → need a mechanism to type unseen NEs!

NER: Knowledge Engineering Approaches

1. Lexical processing

Example sentence: "Norwich Investment Bank plc. today announced ..."

NER: Knowledge Engineering Approaches

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1a. Tokenisation, sentence splitting, morphological analysis, Part-Of-Speech tagging

→ [Norwich]_{NNP} [Investment]_{NNP} [Bank]_{NNP} [plc.]_{NN} [today]_{RB} [announced]_{VBD} ...

NER: Knowledge Engineering Approaches

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1b. Gazetteer Lookup and Tagging:

► **ORG**rganisations, **LOC**ations, **PER**sons, company designators (**CDG**), person titles

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NER: Knowledge Engineering Approaches

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1c. Trigger Word Tagging

▶ **trigger words** allow to classify certain multi-word names

→ Ex.: **Airlines** in "**Wing and Prayer Airlines**"

▶ system has trigger words for **ORG**anisations, **GOV**ernment institutions, **LOC**ations

→ [Norwich]_{NNP/LOC} [Investment]_{NNP} [Bank]_{NNP/ORG-TRIGGER} [plc.]_{NN/CDG} [today]_{RB} [announced]_{VBD} ...

NER: Knowledge Engineering Approaches

2. NE parsing

Hand-produced rules:

- ▶ 177 for proper names
- ▶ 94 for organisation
- ▶ 54 for person
- ▶ 11 for location
- ▶ 18 for time expressions.

NER: Knowledge Engineering Approaches

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A fragment of the proper name grammar:

- ▶ NP $\rightarrow$ ORGAN_NP
- ▶ NAMES_NP $\rightarrow$ NNP
NAMES_NP
- ▶ NAMES_NP $\rightarrow$ NNP
- ▶ ORGAN_NP $\rightarrow$ LIST_LOC_NP
NAMES_NP CDG_NP
- ▶ ORGAN_NP $\rightarrow$ LIST_ORGAN_NP
NAMES_NP CDG_NP
- ▶ ORGAN_NP $\rightarrow$ NAMES_NP '&'
NAMES_NP

NER: Knowledge Engineering Approaches

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NAMES_NP CDG_NP
- ▶ ORGAN_NP → NAMES_NP '&
NAMES_NP

Rule "ORGAN_NP → NAMES_NP '&' NAMES_NP" means:

an unclassified **proper name** (NAMES_NP) followed by '&'
followed by an unclassified **proper name** is an **organisation name**
→ **Marks & Spencer** or **American Telephone & Telegraph**

NER: Knowledge Engineering Approaches

- 3. Discourse interpretation - **Coreference Resolution**
- 3a. When the name class of an **antecedent** (resp. **postcedent**) is known then establishing coreference allows the name class of the **anaphor** (resp. **cataphor**) to be established.

NER: Knowledge Engineering Approaches

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Anaphora/cataphora

In a narrower sense, **anaphora** is the use of an expression that depends specifically upon an **antecedent** expression and thus is contrasted with **cataphora**, which is the use of an expression that depends upon a **postcedent** expression. [Source: Wikipedia]

NER: Knowledge Engineering Approaches

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- ▶ Ex1.: **Ford Motor Co.** was founded in Detroit in 1903. **It** was the first to introduce...

NER: Knowledge Engineering Approaches

3. Discourse interpretation - **Coreference Resolution**

3b. An unclassified NP may be co-referential with a variant form of a classified NP, e.g.:

▶ Ex2.:

- ▶ **Ford Motor Co.** was founded in Detroit in 1903, ..., **Ford** was the first to introduce...
- ▶ **Creative Artists Agency** is a US talent agency ... In 2016, **CAA** had 1,800 employees

- The unclassified NP may be inferred to have the same class as the classified NP.
- Wakao et al. use 45 heuristics of this type for organisation, location, and person names.

NER: Knowledge Engineering Approaches

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3c. An unclassified NP may be co-referential with a definite NP

▶ Ex3.: **Kellogg**, the breakfast cereal manufacturer

NER: Knowledge Engineering Approaches

4. Discourse interpretation - **Semantic Type Inference**

Semantic type information about the arguments in certain **syntactic relations** is used to make inferences permitting the classification of NPs

- 4a. **noun-nous qualification**: NP qualifies an organisation-related object $\rightarrow$ organisation
 $\rightarrow$ Erickson stocks $\Rightarrow$ **[Erickson]**_{ORG}

NER: Knowledge Engineering Approaches

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 $\rightarrow$ Miodrag Jones, president of XYZ $\Rightarrow$ **[Miodrag Jones]**_{PER}

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- 4d. **verbal arguments**: NP names an entity involved in a verbal frame where the semantic type of the argument position is known $\rightarrow$ classify accordingly
 $\rightarrow$ Smith retired from his position as $\Rightarrow$ **[Smith]**_{PER}

NER: Knowledge Engineering Approaches

Evaluation of Wakao et al.

MUC-6 NE evaluation set: a **blind test set** of 30 Wall Street Journal articles containing:

- ▶ 449 organisation names
- ▶ 373 person names
- ▶ 110 location names
- ▶ 111 time expressions

Results:

Proper Name Class	Recall	Precision	F1
Organisation	91 %	91 %	91.0 %
Person	90 %	95 %	92.4 %
Location	88 %	89 %	88.5 %
Time	94 %	97 %	95.5 %
Overall	91 %	93 %	92.0 %

⇒ Best system results on this evaluation had F1 measure = 96.42%

⇒ Human results were 96.68%

NER: Knowledge Engineering Approaches

Strengths

- ▶ **High performance** – only several points behind human annotators
- ▶ **Transparent** – easy to understand what system is doing/why

Weaknesses

- ▶ Porting to another domain requires substantial **rule re-engineering**
- ▶ Acquisition of **domain-specific lexicons**
- ▶ Rule writing requires high **levels of expertise**

Information Extraction: Approaches to NER

Knowledge-engineering

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Supervised learning

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Bootstrapping

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Distant supervision / lightly supervised methods

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NER: Supervised Learning

Aim of Supervised Learning

Address the **portability** / **generalisation** problems inherent in knowledge engineering NER

- ▶ Manually authoring rules $\Rightarrow$ systems learn from **annotated examples**

NER: Supervised Learning

Aim of Supervised Learning

Address the **portability** / **generalisation** problems inherent in knowledge engineering NER

- ▶ Manually authoring rules $\Rightarrow$ systems learn from **annotated examples**
- ▶ Moving to new domain requires only annotated data in the domain
- $\rightarrow$ supplied by **domain expert** without need for expert computational linguist

NER: Supervised Learning

Aim of Supervised Learning

Address the **portability** / **generalisation** problems inherent in knowledge engineering NER

A wide variety of supervised learning techniques have been tried, including:

- ▶ Hidden Markov Models (HMM)
- ▶ Decision Trees
- ▶ Maximum Entropy models
- ▶ Support Vector Machines (SVM)
- ▶ Conditional Random Fields (CRF)
- ▶ AdaBoost
- ▶ Deep Learning

NER: Supervised Learning

Two types of systems that may learn:

1. **patterns** that match extraction targets → less developed recently
 2. **classifiers** that label tokens as **beginning/inside/outside** a **tag type**
- Systems operate as **Sequence Labelling** systems

NER: Supervised Learning

In **sequence labelling**, each token is given one of three label types **B**, **I** or **O**:

- ▶ **B**_{CLASS} if the token is at the **beginning** of a named entity of class = *CLASS*
 - ▶ where $CLASS \in \{\mathbf{ORG}, \mathbf{PER}, \mathbf{LOC}, \text{etc.}\}$
- ▶ **I**_{CLASS} if the token is **inside** a named entity
- ▶ **O** if the token is **outside** any named entity

⇒ referred to as the **BIO** scheme

NER: Supervised Learning

Example:

[American Airlines]_{ORG}, a unit of [AMR Corp.]_{ORG}, immediately matched the move, spokesman [Tim Wagner]_{PER} said.

[Source: Jurafsky and Martin, 2nd ed., p. 730]

NER: Supervised Learning

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In **BIO** encoding this example looks like this:

American	B _{ORG}
Airlines	I _{ORG}
a	O
unit	O
of	O
AMR	B _{ORG}
Corp.	I _{ORG}
,	O
immediately	O
matched	O
the	O
move	O
,	O
spokesman	O
Tim	B _{PER}
Wagner	I _{PER}
said	O
.	O

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Supervised ML technique:

- ▶ Trained on a corpus of labelled sequences, like this
- ▶ Aim: predict the labelling of a new, unlabelled example.



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Airlines	I _{ORG}
a	O
unit	O
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AMR	B _{ORG}
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,	O
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matched	O
the	O
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,	O
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Tim	B _{PER}
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said	O
.	O

NER: Supervised Learning

Features for Sequence Labelling

Each token is typically represented as a set of features:

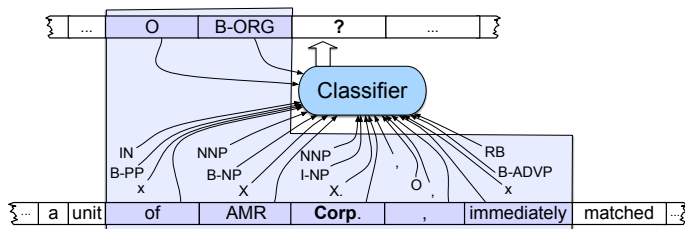
- ▶ the **token** itself
- ▶ characteristics of the **token**: stem, Part-Of-Speech, etc.
- ▶ characteristics of the **neighbouring tokens** (window of ± 2 or 3 tokens either side)

Typical features used: [Source: Jurafsky and Martin, 2nd ed., p. 731]

Features	Explanations	Ex.: "The shop L'Occitane en Provence"
Lexical items	The token to be labeled	L'Occitane
Stemmed lexical items	Stemmed version of the target token	Occitan
Shape	Orthographic pattern of the target token	X'Xxxxxxx or X'Xx
Character affixes	Character-level affixes	L, L', L'O, L'Oc, tane, ane, ne, e
Part of Speech	Part of speech of the token	NNP
Syntactic chunk labels	Base-phrase chunk label	B-NP
Gazetteer or name list	Token seen in a named entity lists	-
Predicted token(s)	Previously predicted classes	O O
Bag of words/Bag of n-grams	Tokens and/or n-grams in the context	The, shop, The shop , etc.

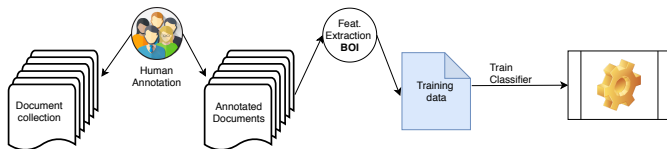
NER: Supervised Learning

Graphical representation of features used to NER:

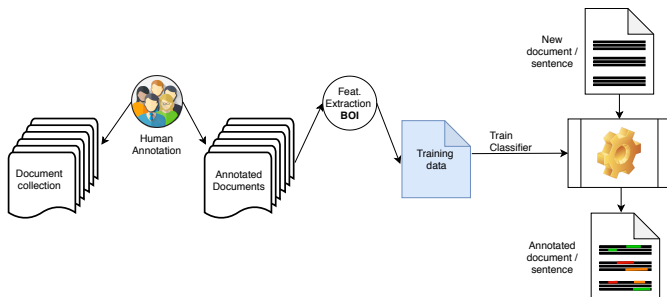


[Source: Jurafsky and Martin, 2nd ed., p. 733]

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BIO sequence labelling: Carreras et al. (2003)

[Carreras et al., 2003]

→ best score at CONLL 2003 NER shared task

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Use of a two-pass method:

1. **NE detection:** assign un-typed **BIO** tags → find NE boundaries **whatever the class**
2. **NE classification:** assign a class to all **BI** tags

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- ▶ Type pattern of consecutive word in context: functional (f), capitalized (C), lowercased (l) etc. → "John Smith payed 3 euros" ⇒ CClxl

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Use of **Adaboost** classifier:

- ▶ **Adaptive Boosting** is a meta-algorithm that combines outputs of other learning algorithms (called **weak learners**). Subsequent weak learners are tweaked in favor of the instances that are misclassified by previous classifiers.

NER: Supervised Learning

BIO sequence labelling: Carreras et al. (2003)

[Carreras et al., 2003]

	English		German	
	Precision (%)	Recall (%)	Precision (%)	Recall (%)
NE Detection	91.93	94.02	85.85	72.61
NE Classification	84.05	85.96	75.47	63.82
	Accuracy		Accuracy	
Oracle	95.14		85.14	

Oracle: Classification with a perfect Detection

⇒ LOC and PER score consistently higher than ORG and MISC

Information Extraction: NER recap → Supervised Learning

Example: apply the BIO annotation

Despite stars Kevin Hart and Dwayne 'The Rock' Johnson having very busy schedules, producer Hiram Garcia has hinted that things are already underway in Hollywood.

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Despite stars [Kevin Hart]_{PER} and [Dwayne 'The Rock' Johnson]_{PER} having very busy schedules, producer [Hiram Garcia]_{PER} has hinted that things are already underway in [Hollywood]_{LOC}.

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[Despite]_O [stars]_O [Kevin]_{B_PER} [Hart]_{I_PER} [and]_O [Dwayne]_{B_PER} [']_{I_PER} [The]_{I_PER} [Rock]_{I_PER} [']_{I_PER} [Johnson]_{I_PER} [having]_O [very]_O [busy]_O [schedules]_O [,]_O [producer]_O [Hiram]_{B_PER} [Garcia]_{I_PER} [has]_O [hinted]_O [that]_O [things]_O [are]_O [already]_O [underway]_O [in]_O [Hollywood]_{B_LOC}.

Information Extraction: NER recap → Supervised Learning

Example: apply the BIO annotation

William MacDonald took the bus from Kelham Island to Manor Park, aiming to visit the Sheffield Manor Lodge.

Information Extraction: NER recap → Supervised Learning

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Information Extraction: NER recap → Supervised Learning

Example: apply the BIO annotation

William MacDonald took the bus from Kelham Island to Manor Park, aiming to visit the Sheffield Manor Lodge.

[William MacDonald]_{PER} took the bus from [Kelham Island]_{LOC} to [Manor Park]_{LOC}, aiming to visit the [Sheffield Manor Lodge]_{LOC}.

[William]_{B_PER} [MacDonald]_{I_PER} [took]_O [the]_O [bus]_O [from]_O [Kelham]_{B_LOC} [Island]_{I_LOC} [to]_O [Manor]_{B_PER} [Park]_{I_LOC} [,]_O [aiming]_O [to]_O [visit]_O [the]_O [Sheffield]_{B_LOC} [Manor]_{I_LOC} [Lodge]_{I_LOC}.

Information Extraction: Overview

- ▶ **Introduction to Information Extraction**
 - ▶ Definition + contrast with IR
 - ▶ Example Applications
 - ▶ Overview of Tasks
 - ▶ Overview of Approaches
 - ▶ Evaluation + Shared Task Challenges
 - ▶ Brief(est) history of IE
- ▶ **Named Entity Recognition**
 - ▶ Task
 - ▶ Approaches: Rule-based, Supervised Learning
 - ▶ **Entity Linking**
- ▶ **Relation Extraction**
 - ▶ Task
 - ▶ Approaches: Rule-based, Supervised Learning, Bootstrapping, Distant Supervision

NER: Entity Linking

IE using knowledge base population (**KBP**)

- ▶ facts are gathered from open access web sources
- ▶ then used to build a structured information repository

⇒ To correctly assemble facts, the entities must be linked to the appropriately entry in the KB

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Entity Linking Task:

Given a text with a **recognised NE mention** in that text and a **knowledge base (KB)**, such as Wikipedia, link the NEs to the matching entry in the KB if there is one, else create an entry.

NER: Entity Linking

Is this task difficult? – yes!!

- Wikipedia contains over 200 entries for **John Smith**, at least 1,716 places called **San José** (or San Jose)

[Source: Wikipedia]

Explorer [[edit](#)]

- John Smith (explorer) (1580–1681), helped found the Virginia Colony and became Colonial Governor of Virginia

Academics [[edit](#)]

- John Smith (anatomist and chemist) (1721–1797), professor of anatomy and chemistry at the University of Oxford, 1768–97
- John Blair Smith (1764–1799), president of Union College, New York
- John Smith (Cambridge, 1768), vice chancellor of the University of Cambridge, 1768 until 1767
- John Smith (astronomer) (1774–1805), Lucasian Professor of Astronomy and Master of Caius
- John Smith (zooarchaeologist) (died 1920), professor of languages at Dartmouth College
- John Augustine Smith (1780–1866), president of the College of William and Mary, 1814–1826
- John Smith (botanist) (1780–1800), curator of Kew Gardens
- John Smith (physician) (c.1600–1676), Scottish physician specialising in treating the insane
- John Alexander Smith (physician) (1816–1883), Scottish physician, antiquarian, archaeologist and ornithologist
- J. Lawrence Smith (1816–1883), American doctor and chemist
- John Smith (dentist) (1825–1910), founder of Edinburgh's School of Dentistry
- John Campbell Smith (1828–1914), Scottish writer, advocate and Sheriff-Substitute of Forfarshire
- John Donnell Smith (1829–1908), biologist and taxonomist
- John McCarville Smith (1844–1918), Australian metallurgist and bacteriologist
- John Alexander Smith (1863–1909), British bioidal philosopher
- John Maynard Smith (1920–2004), geneticist
- John Cyril Smith (1922–2002), leading authority on English criminal law
- John Derek Smith (1924–2005), Cambridge molecular biologist
- John Smith (sociologist) (1927–2002), English sociologist
- John D. Smith (born 1946), biologist at the University of Cambridge
- John H. Smith (mathematician), Boston College educator (retired 2005)

Arts [[edit](#)]

- John Smith (engraver) (1652–1742), English mezzotint engraver
- John Smith (English poet) (1662–1717), English poet and playwright
- John Christopher Smith (1719–1746), English composer
- John Warwick Smith (1746–1801), British watercolour landscape painter and illustrator
- John Stafford Smith (1750–1836), composer of the tune for "The Star-Spangled Banner"
- John Raphael Smith (1751–1812), English mezzotint engraver and painter
- John Thomas Smith (engraver) (1756–1832), draughtsman, engraver and antiquarian
- John Smith (clockmaker) (1770–1816), Scottish clockmaker
- John Rubens Smith (1770–1849), London born painter, printmaker and art instructor who worked in the United States
- John Smith (architect) (1781–1862), Scottish architect
- John Smith (art historian) (1781–1855), British art dealer
- John Otin Smith (1790–1843), English wood engraver
- John Frederick Smith (1800–1860), English novelist
- John Rowson Smith (1810–1866), American seascene painter
- John G. Smith (poet) (died 1881), Scottish poet
- John Myer Smith (1830–1912), British artist and designer
- John Berryman (1914–1972), originally John Allyn Smith, American poet
- John Smith (Canadian poet) (born 1927), Canadian poet
- John Smith (actor) (1924–1990), American actor
- John H. Smith (born 1942), Canadian film director and screenwriter
- John Templeton Smith (born 1943), British-born American thriller novelist
- John Smith (English filmmaker) (born 1952), avant-garde filmmaker
- John Warner Smith (born 1952), American poet and educator
- John P. Smith, American soap opera writer
- John Smith (comic writer) (born 1967), British comic writer
- John Smith (musician), English contemporary folk musician and recording artist

Religion [[edit](#)]

- John Smith (Bishop of Landaff) (died 1478), Welsh bishop
- John Smith (Psalterist) (1618–1650), one of the founders of the Cambridge Psalterists
- John Smith (Bishop of Kilkee and Achonry) (died 1686), Irish Anglican priest
- John Smith (Unitarian) (fl. 1648–1725), Unitarian writer
- John Smith (priest, born 1660) (1660–1710), English writer of Bells
- John Smith (uncle of Joseph Smith) (1781–1854), Presiding Patriarch and member of the First Presidency of The Church of Jesus Christ of Latter-day Saints
- John Smith (Restoration Movement) (1784–1808), early Restoration Movement leader
- John Smith (missionary) (1790–1846), English missionary to Denmark
- John Smith (Revivalist) (1794–1831), English Methodist minister known as "The Revivalist"
- John Smith (Anglican priest) (1798–1870), first person to transcribe the Diary of Samuel Pepys
- John Smith (nephew of Joseph Smith) (1802–1811), Presiding Patriarch of The Church of Jesus Christ of Latter-day Saints
- John Henry Smith (1804–1811), apostle in "The Church of Jesus Christ of Latter-day Saints"
- John Taylor Smith (1800–1838), Anglican Bishop of Sierra Leone
- John Smith (Archdeacon of Wilton) (1820–2002), Anglican priest
- John H. Smith (priest) (1825–2016), American Roman Catholic bishop
- John H. Smith (bishop) (1904–2012), American Episcopal priest, Bishop of the Episcopal Diocese of West Virginia, 1969–1989
- John Smith (Buck's Squid) (died 2018), Australian founder of the Buck's Squid enterprise club

Sports [[edit](#)]

Cricket [[edit](#)]

- John Smith (Essexfield cricketer), English cricketer
- John Smith (cricketer, born 1833) (1833–1906), Lancashire and Yorkshire cricketer
- John Smith (cricketer, born 1834) (1834–?), Nottinghamshire and All-England Eleven cricketer
- John Smith (cricketer, born 1835) (1835–1888), English cricketer
- John Smith (Derbyshire cricketer) (1841–1888), English cricketer, played for Derbyshire
- John Smith (cricketer, born 1843) (1843–1873), English cricketer, played for Cambridgeshire 1862–72
- John Smith (cricketer, born 1860) (1860–1948), English cricketer, played for Leicestershire 1901
- John Smith (cricketer, born 1906) (1906–1971), English cricketer, played for Leicestershire 1930–1959
- John Smith (cricketer, born 1936) (born 1936), Australian cricketer
- John Smith (New Zealand cricketer) (born 1965), New Zealand cricketer also known as Campbell Smith

Association football [[edit](#)]

- John Smith (footballer, born 1855) (1855–1934), Scottish footballer of the 1870s and 1880s
- John Smith (footballer, born 1862) (1862–1911), Scottish footballer who played as a winger
- John Smith (footballer, born 1866) (1866–?), Scottish footballer (for United, Manchester, Scotland)
- John Smith (footballer, born 1907), English footballer for Bradford City and Blackburn Rovers
- John Smith (footballer, born 1921) (1921–?), English footballer
- John Smith (footballer, born 1927) (1927–2005), English footballer (Leicester)
- John Smith (footballer, born 1928) (1928–1986), West Ham United
- John Smith (footballer, born 1944), Welsh footballer (Sheffrham)
- John Smith (footballer, born 1973), football full back (Thames Valley)
- John Smith (footballer, born 1975), retired English footballer

Criminals [[edit](#)]

- John Smith (housebreaker) (1661–after 1727), London housebreaker who evaded hanging three and was eventually transported to Virginia
- John Smith (murderer) (born 1951), American convicted murderer who killed his first and second wives
- John Eldon Smith (1830–1883), convicted of the murders of Ronald and Juana Nola

Businessmen [[edit](#)]

- John Smith (brewer) (1804–1876), Tadcaster brewery founder in North Yorkshire, UK
- John F. Smith, Jr. (born 1938), former chairman and chief executive officer, General Motors
- John L. Smith (1923–1990), African American abolitionist, Underground Railroad confidant and politician
- John L. Smith (pharmaceutical executive) (1889–1990), German-born American chemist, pharmaceutical executive and Major League Baseball team owner
- John Smith (executive) (born 1967), former chief executive officer, BBC Worldwide Ltd
- John Russell Smith (1810–1894), English bookbinder and bibliographer

Others [[edit](#)]

- John Smith (Chippewa Indian) (died 1902), reputed to have died at the age of 137

NER: Entity Linking: Approaches

- ▶ Simple idea: given a text T containing an NE mention m and using Wikipedia as a KB
 1. index all pages in the KB using IR
 2. build a query from T containing m + search the KB
 3. pick the best ranked page from step 2
- does not work very well

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 1. index all pages in the KB using IR
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- does not work very well
- ▶ More successful approaches consider disambiguating all NEs jointly
 - ▶ **Intuition:** in disambiguating a text mentioning **Ashoka** and **Sheffield**, the **Ashoka** mentioned is likely to be in Sheffield, while the Sheffield is likely to be one containing an Ashoka restaurant.
 - ▶ See, e.g., Alhelbawy and Gaizauskas (2014)

NER: Conclusion

Named Entity Recognition (NER) is a core IE technology

- ▶ detect and classify all mentions of NE of a given set of entity types within a given text
- ▶ now relatively mature and at “usable” performance levels

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Open challenges include:

- ▶ reducing the amount of training data needed via, e.g. bootstrapping techniques
- ▶ exploiting existing structured data sources to generate “weakly labelled” training data (aka distant supervision)
- ▶ expanding the classes of entities addressed
- ▶ developing NERs for languages other than English

NER: Conclusion

GATE platform has multiple NER tools

- ▶ support for multiple domains (e.g. Twitter, Biomedical)
- ▶ multiple languages (e.g. English, Russian, German, Welsh)

<https://cloud.gate.ac.uk>

Reading material

- ▶ Daniel Jurafsky & James H. Martin (2021).
Chapter 17: Information Extraction.
Speech and Language Processing, 3rd ed. draft.
<https://web.stanford.edu/~jurafsky/slp3/17.pdf>

Credit: slides based on Dr Loïc Barrault's slides for the autumn 2020/2021 course.

References I

Carreras, X., Màrquez, L., and Padró, L. (2003).

A Simple Named Entity Extractor Using AdaBoost.

In *Proceedings of the Seventh Conference on Natural Language Learning at HLT-NAACL 2003 - Volume 4*, CONLL '03, pages 152–155, Stroudsburg, PA, USA. Association for Computational Linguistics.

Cunningham, H., Maynard, D., Bontcheva, K., Tablan, V., Aswani, N., Roberts, I., Gorrell, G., Funk, A., Roberts, A., Damljjanovic, D., Heitz, T., Greenwood, M., Saggion, H., Petrak, J., Li, Y., Peters, W., and Derczynski, L. (2014).

Developing language processing components with GATE Version 8.

University of Sheffield Department of Computer Science.

Wakao, T., Gaizauskas, R., and Wilks, Y. (1996).

Evaluation of an algorithm for the recognition and classification of proper names.

In *The 16th International Conference on Computational Linguistics, COLING*, volume 1.